Woongjib Choi

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Research Interests

Audio and Speech Signal Processing, Audio Coding, Digital Signal Processing, Generative Models

Education

Massachusetts Institute of Technology Sep. 2026 – Present
Ph.D. Student in Electrical Engineering and Computer Science Cambridge, MA, USA
- Advisor: Prof. Paris Smaragdis

Yonsei University Mar. 2024 – Feb. 2026
M.S. in Electrical and Electronic Engineering Seoul, South Korea
- Advisor: Prof. Hong-Goo Kang
- GPA: 4.08 / 4.30

Yonsei University Mar. 2018 – Feb. 2024
B.S. in Electrical and Electronic Engineering Seoul, South Korea
- Includes two-year mandatory military service (Apr. 2020 – Oct. 2021)
- GPA: 3.83 / 4.30

Research Experience

Graduate Researcher Sep. 2026 – Present
Research Laboratory of Electronics, Massachusetts Institute of Technology Cambridge, MA, USA
- Researching **representation learning** and **time-frequency** analysis for audio and speech.

Postgraduate Researcher Mar. 2026 – May 2026
DSPAI Lab, Yonsei University Seoul, South Korea
- Conducted research on **neural audio codecs**, **tokenization**, and **generative models** for audio.
- Collaborated with *Prof. Yongshik Lee* on vehicle state classification from unintentional electromagnetic emissions, including STFT-based signal analysis and DNN-based classification.

Graduate Researcher Mar. 2024 – Feb. 2026
DSPAI Lab, Yonsei University Seoul, South Korea
- Developed **bandwidth extension** methods via **flow matching** and **self-supervised representations**.
- Led the development of GAN-based post-processing for legacy **audio codecs** in the **ETRI-Yonsei Joint Research Project** and served as project leader for the final six months, resulting in 1 international conference paper, 1 domestic paper, and 1 patent.

Undergraduate Research Intern Jun. 2023 – Jan. 2024
SRIP Lab, Yonsei University (Supervised by Prof. Moon Gi Kang) Seoul, South Korea
- Implemented classical and deep learning-based image processing algorithms, including white balancing, denoising, demosaicing, and super-resolution.
- Undergraduate thesis: *A Technical Overview of Deep Learning-Based Single-Image Super-Resolution Methods*

Publications

Peer-Reviewed Publications

- [1] Sangmin Lee, Woojin Chung, **Woongjib Choi**, and Hong-Goo Kang, "MoLGE: Mixture of Language Group Experts for Efficient Scaling of Massively Multilingual Speech Recognition," in ***Conference on Language Modeling (COLM)***, 2026. [paper]
- [2] Sangmin Lee, Eekgyun Ahn, **Woongjib Choi**, and Hong-Goo Kang, "UR-BERT: Scaling Text Encoders for Massively Multilingual TTS Through Universal Romanization and Speech Token Prediction," in ***INTERSPEECH***, 2026. [paper]

- [3] **Woongjib Choi**, Sangmin Lee, Hyungseob Lim, and Hong-Goo Kang, "UniverSR: Unified and Versatile Audio Super-Resolution via Vocoder-Free Flow Matching," in ***ICASSP***, 2026. [paper] [demo] [code]
- [4] **Woongjib Choi**, Byeong Hyeon Kim, Hyungseob Lim, Inseon Jang, and Hong-Goo Kang, "Neural Spectral Band Generation for Audio Coding," in ***INTERSPEECH***, 2025. [paper]

Preprints & Submissions

- [5] Sangmin Lee, **Woongjib Choi**, Jihyun Kim, and Hong-Goo Kang, "SAGE-LD: Towards Scalable and Generalizable End-to-End Language Diarization via Simulated Data Augmentation," *arXiv preprint*, 2025. [paper]

Domestic Publications

- [6] Jaehun Shin, **Woongjib Choi**, Byeong Hyeon Kim, Inseon Jang, and Hong-Goo Kang, "DNN-based Speech Codec Enhancement Utilizing Conditional Flow Matching," in *Proc. KIBME Conf.*, 2025.
- [7] **Woongjib Choi**, Byeong Hyeon Kim, and Hong-Goo Kang, "Non-Blind Speech Bandwidth Extension Using Pre-Trained Self-Supervised Features," in *Proc. IEIE Conf.*, 2024.

Patents

Encoder and Decoder for Audio Coding Using Side Information Generated from Original Signal and Method of Operation Thereof — KR 10-2025-0096305

Teaching Experience

Teaching Assistant — Mar. 2024 – Jun. 2025
Department of Electrical and Electronic Engineering, Yonsei University — Seoul, South Korea

- **EEE2111: Analog Electronics Lab** (Spring 2025)
- **EEE2060: Signals and Systems** (Fall 2024)
- **EEE4610: Capstone Design** (Fall 2024)
  - Advised project on unified speech enhancement bridging personalized and unconditional models.
- **EEE4420: Digital Signal Processing** (Spring 2024)

Awards & Honors

Dean's List, Yonsei University (top 3% of college) — Spring 2023

Academic Merit Scholarship, Yonsei University — Spring & Fall 2023

Skills

Programming: Python, MATLAB, C/C++, Verilog · **Frameworks:** PyTorch, Hugging Face, Librosa, SciPy
Tools: Linux, Git, Slurm, Arduino, Xilinx Vitis/Vivado · **Languages:** Korean (native), English (advanced)

Extracurricular Activities

Editor-in-Chief, Humans of Yonsei (Student Storytelling and Interview Organization) — Mar. 2022 – Jun. 2023

Sergeant, Republic of Korea Army (Repair Parts and Tool Supply Specialist) — Apr. 2020 – Oct. 2021

Freshman Class Representative, Dept. of EEE, Yonsei University — Mar. 2018 – Sep. 2018